Biometric Systems Can Achieve Better Security: New Optimal Operating Point For Biometric Credentials

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Abstract

Credentials are a fundamental component in securing users' digital assets. Biometric credentials such as fingerprints and face recognition are essential for modern authentication, as, unlike passwords, they are intrinsically bound to the user, making them harder to steal, reuse, or forget. However, by artificially reproducing biometric features using AI, attackers can impersonate users and gain access to their digital assets more easily, urging the need for more robust and secure user authentication. Today, many modern biometric sensors rely on a predetermined threshold and return only a binary decision (True or False) during a scan. Since the evaluation is performed inside a Trusted Execution Environment (TEE), the operating point cannot be externally adjusted, which may be suboptimal for different users and increase vulnerability to spoofing attacks.

In this work, we explore how biometrics along with other credentials can be fused to form a robust authentication mechanism for protecting users' digital assets.

We propose a dynamic operating point that achieves a higher probability of success than the commonly used operating point in biometric systems. Through numerical analysis and first- and second-order approximations of different attacker distributions, we symbolically derive the optimal operating point for several credential combinations, including biometric credentials alone or in multi-credential mechanisms. We quantify how the attacker distribution affects the resulting improvement in the mechanism's security.

To support our theoretical claims, we evaluate its effectiveness on real biometric data and characterize the conditions under which integrating a biometric credential enhances the mechanism's security. We prove that any asymptotically separable biometric credential can enhance the security of a mechanism based on binary credentials. We show measurements demonstrating that such an improvement can decrease the failure of the system by orders of magnitude under spoofing attacks.

Our findings have immediate and practical implications for the design and improvement of authentication mechanisms.

1 Introduction

Authentication is a critical component of security, serving as the gatekeeper that verifies the identity of users and grants access to digital assets [4, 14, 23]. As advances in artificial intelligence increasingly enable attackers to impersonate more easily, companies are turning to adopt more advanced authentication mechanisms. For example, passkeys, which are based public-key cryptography and biometrics, are emerging as a replacement or second factor for passwords [12]. This type of authentication can be modeled by a continuous distribution of the credential score.

In contrast, other credentials such as passwords, OTPs and physical devices follow a discrete distribution, since the authentication process results in a strict binary decision.

Biometric credentials (e.g, fingerprints, face recognition or iris) rely on physical biometric features which are not evaluated in a discrete way. Rather, they are measured by a mechanism that determines whether there are enough similarities above a predetermined threshold between an input and a stored figure. Nevertheless, identity theft incidents are frequent [7, 17], as the global biometric market, valued at $34B, has become a preminent tarket for attackers. With deepfake fraud now causing average losses of $680,000 per incident at large enterprises [24], the need for more robust and secure authentication mechanisms is more critical than ever.

Previous work (§2) examined various biometric authentication methods and their performance. Lin Hong et al. [9] study and build an integrated system made of multiple biometric credentials. Prabhakar et alx . [19] and Sarkar et al. [22] examined the best way to use these credentials and the tradeoffs between different operating points, highlighting its impact on both the accuracy and security of biometric systems. Other works were focused on improving scoring algorithms, to increase the separability between user and attacker distributions [5,8,10,13,15,18,26]. However, these algorithms were neither accurate enough, nor robust against emerging AI-driven challanges, prompting companies and organizations to the development of complexed authentication mechanisms such as multifactor authentication (MFA) and interactive protocols [3, 14]. Eyal et al. [6] designed and formulated a foundational wallet model, defining four possible states for each credential – safe, loss, leak and theft. However, this work did not address the type of credentials involved, nor proposed a method to determine the probability of each credential's state.

In this work, we study the integration and usage of biometric credentials in context of the wallet model. We looked for a dynamic and optimal operating point that yields a higher probability of success, addressing the challanges of spoofing attacks. First, we define the authentication model (§3) built on a set of credentials $\{c_1, c_2, \ldots, c_n\}$ distributed either discretly or continuously. The mechanism interacts with 2 players: user and attacker, trying to identify the user based on the provided credentials by the player. We follow the definitions given by Eyal [6], which define a wallet as a predicate of availability of N keys, where each key can be in one of the following states: safe, loss, leak and theft. Then, we associate a probability with each one of the states, and define the probability of success of a wallet depends on the scenario - which credential each player knows. For example, continuos credential is evaluated by a mechanism that outputs a confidence score for both players. This score characterizes the geniune and imposter distributions, which are used to establish two error rates: false acceptance rate (FAR) and false rejection rate (FRR).

Analyzing the effect of a credential on the success probability, led us to start with a wallet comprised of a single biometirc credential (§4). We examined distribution functions which are first-and second-order approximation of a biometric credential, trying to find a dynamic and optimal point that increase the success probability while balancing the two error rates. We symbolically calculated the optimal point and found that it is different than the commonly used one, known as the equal error rate (EER) [22]. For example, a first order apporximation of the biometric credential which is uniformly distributed, yields an optimal point at $T = \frac{a_1+u_2}{2}$, while the EER point is at $T = \frac{a_1 \cdot u_1 - a_2 \cdot u_2}{a_1 - a_2 + u_1 - u_2}$, where a_1, a_2 and u_1, u_2 are the bounds of the attacker and user distributions respectively. This suggests a different and dynamic operating point which depends on the user characteristics and the attacker's capabilities.

As each credential can influence the wallet's success probability differently, a more complex and realistic case was examined, where a wallet is comprised of multiple and different credentials (§5). Our analysis revealed that the optimal operating point for a wallet comprised of multiple biometric credentials is not necessarily the same as the optimal point for each individual credential (§5.1). Using numeircal optimization algorithm, lead us to observe that there is

about 6% improvement in the success probability between the EER and the optimal operating point.

We find that the optimal point of a wallet comprised of a single discrete credential and a second order approximation of the biometric credential which is quadradically distributed, differs from the EER as well. We link between the values of the discrete credential and the influence it has on the operating point of the biometric credential. Our findings show that high safe values and low leak values of the discrete credential shifts the operating point towards the lower boundry of the biometric credential, making the success of the wallet becomes mainly dependent on the discrete credential, vise versa. We observe that the integration of the biometric credential strictly improves the security of the wallet compared to using the discrete credential alone, of every discrete credential with non zero state's probability of safe and loss or leak (§5.2). Thus, the question arises: is that improvement consistent across all biometirc credentials?

To answer this question, we formalize four proposotions that characterize the conditions under which integrating a biometric credential into a wallet composed of a single discrete credential, enhances its security (§6). We show where no security gain is achieved when integrating biometric credential with discrete one (§6.1). We find that the improvement is consistent across all *asymptotic separabile* biometric credentials which is necessary and sufficient condition, as long as the discrete credential has a non zero state's probability of safe and loss or leak (§6.2). This finding has immediate and practical implications for the design and improvement of authentication mechanisms, suggesting that integrating biometric credentials can enhance security in a wide range of scenarios. To provide more intuiutive and practical condition, we define *asymptotic dominancy*, which is a sufficient condition for the enhancement of the wallet's success. This condition means that the attacker's distribution is dominated by the user's distribution in the limits where the matching score s goes to infinity and vise versa (§6.3).

To support our theoretical findings, we conduct an empirical analysis using real biometric data and sourceafsi, an open source algorithm [25] (§7). Firstly, we use dataset of different users and evaluating the matching score between different users as the imposter distribution and different tamplates of the same user as the geniuine. The integration of this fingerprint biometric credential into a wallet composed of a single discrete credential, consistently enhance its security, confirming our theoretical predictions. However, the improvement was not significant due to high seperability between the 2 players. We further conduct a simulation of a spoofing attack, where an adversary generates fingerprint samples that impersonate the legitimate user. These samples are characterized by matching scores closely resemble those of the user, taken from LivDet database [8, 15, 18, 26]. In this case, we observe a significant improvement in the wallet's success probability when integrating the fingerprint credential using our method, demonstrating the practical implications of our theoretical findings.

Our contributions are both theoretical and practical. We formalized the user's success probability of a wallet composed of biometric credentials. We found a different and dynamic operating point that yields a higher success probability. We analyzed the integration of a biometric credential into a wallet composed of multiple heterogeneous credentials, and characterized the conditions under which this integration enhances the wallet's security. We proved that any biometric credential under a specified condition always enhances the user's success prbability. We further supported our theoretical findings with empirical analysis using real biometric data, demonstrating the practical implications of our work for the design and improvement of authentication mechanisms. We conclude (§8) by reviewing an open direction related to the design and implementation of authentication mechanisms, built on our suggestions. More specifically, we suggest a mechanism that can dynamically adjust the operating point and stricting the threshold as the number of authentication attempts increases, to mitigate the risk of spoofing attacks.

2 Related Work

Authentication continuous to be an active and prominent research area, as the need for robust and secure authentication mechanisms becomes increasingly critical in the emerging artificial intelligence landscape. This research field had been widely explored, covering large variety of topics such as credentials, algorithms and mechanisms. Credentials are the core of the authentication process. Some credentials rely on exclusive access, such as cryptographic signing keys, passwords [16], OTPs [20], or physical devices. These credentials can be viewed as discretely distributed, since authentication results in a strict binary decision. Other credentials rely on biometric features such as fingerprints, facial recognition, or iris scans [11]. Such credentials can be described by a continuous distribution, as each scan produces a confidence score that the system evaluates against a threshold. Some works on authentication mechanisms, were focused on imrpoving scoring algorithms [5, 8, 10, 13, 15, 18, 26], to increase the separability between user and attacker distributions. However, these algorithms were neither accurate enough, nor robust against spoofing attacks, prompting companies and organizations to the development of complexed authentication mechanisms such as multifactor authentication (MFA) and interactive protocols [3, 14]. Our focus is on the integration of biometric credentials, along with other credentials into authentication mechanisms. Particularly, we aim to improve the probability of users successfully defending their digital assets, by integrating biometric credentials, and suggest a different way to use them.

Lin Hong et al. [9] found that automatic personal identification system based solely on fingerprints or faces is often not able to reach performance requirements. They integrated a biometric system which makes personal identification by integrating both faces and fingerprints operates during identification mode. The decision fusion of their system is designed to operate at measurement level, that is the system doesn't just output a single decision or label but rather a set of labels with confidence values. The system decision is based on the confidence of each one of the modules which might lead to a more reliable and informative overall confidence score.

These confidence values are characterized by the FAR values of the credential. The FAR values were calculated as the number of similarities between 2 measures above a threshold t. If there are 5 similarities out of 8 between 2 fingerprints and the threshold was 4, these 2 fingerprints will be considered equal.

Prabhakar et al. [19] examined the tradeoff between False Acceptance Rate (FAR) and False Rejection Rate (FRR), highlighting its impact on both the accuracy and security of biometric systems. Their findings suggest that most applications aim to operate at a point that balances these two metrics.

This concept is further developed by Sarkar et al. [22], who introduced it as the Equal Error Rate (EER). Sarkar defined this point as the operating point of the biometric system, emphasizing that the lower the EER value, the greater the performance of the biometric authentication system. However, none of these works defined a success function that characterizes the security of the system or its probability of successful authentication. Moreover, they did not explain how to integrate a biometric credential into an authentication mechanism composed of multiple heterogeneous credentials, nor analyze its efficacy.

Eyal et al. [6] designed and formulated a foundational wallet model, defining four possible states for each credential – safe, loss, leak and theft. However, this work did not address the type of credentials involved, nor proposed a method to determine the probability of each credential's state.

In this work, we present a success probability of a wallet, and suggest a different and dynamic operating point. We further analyze the integration of a continuous credential into a wallet composed of multiple heterogeneous credentials, and characterize the conditions under

which this integration enhances the wallet's security. Our findings can be applied to the design and imrpovement of authentication mechanisms, particularly in the context of biometric authentication and SSO mechanisms that rely on behavioral or environmental signals.

3 Authentication Model

An authentication mechanism M is built on a set of credentials $\{c_1, c_2, \ldots, c_n\}$ and two players: a user U and an attacker A. Both players, try to satisfy the system with their own set of credentials $\{c_1, c_2, \ldots, c_n\}$ in order to reach the asset. Each credential c_i can be either discrete or continuously distributed. In this paper, we will focus on continuously distributed credentials. We will formalize the systems success and identify the optimal operating point for both single and multiple credentials.

3.1 Model Details

We follow the definitions given by Eyal [6]. A wallet w is defined by predicate of availability of N keys. Each key can be in one of the following states. Safe: only the user has access to the key. Loss: neither the user nor the attacker has access to the key. Leak: both the user and the attacker have access to the key. Theft: only the attacker has access to the key.

We define the probability associated with each state as follows: P_{safe}, P_{loss}, P_{leak} and P_{theft} corresponding respectively to the states of safe, loss, leak, and theft. Furthermore, the confidence associated with different decisions may be characterized by the genuine distribution and the impostor distribution, which are used to establish two error rates: false acceptance rate (FAR), which is defined as the probability of an impostor being accepted as a genuine individual and false rejection rate (FRR), which is defined as the probability of a genuine individual being rejected as an impostor.

A wallet w is comprised of a set of credentials $\{c_1, c_2, \ldots, c_n\}$ clasified as one of previously defined states.

Denoted S, i.e., $\{c_1, c_2, \ldots, c_n\} \in S = \{\text{safe}, \text{loss}, \text{leak}, \text{theft}\}$ the state of each credential. A scenario σ is defined as a vector of states of each credential in the wallet. Denote σ_i the state of credential c_i in scenario σ. An availability vector represents the availablity of all credentials to a user U or an attacker A. Denote by σ^U,σ^A the availability vector of the user and the attacker respectively.

In the following sections we will represent all biometric credentials as the probability function of the matching score s denoted as $P_U[s]$. We follow the definitions given by Jain et al. [11] and generalize it to all biometric credentials. If the stored reference template of the user U is represented by X_U and the acquired input for recognition is represented by X_Q then:

H_0 input X_Q does not come from the same person as the reference template X_U.

H_1 input X_Q comes from the same person as the reference template X_U.

D_O, D_1 are the imposter and geniuine persons trying to authenticate into the account. The matching score s, is usually a single number, that quantifies the similarity between an input X_Q and and the reference store template in the database X_U [19]. If the matching score $s(X_Q, X_U)$ is above a threshold t then decide D_1 else, decide D_0.

4 Single Credential

For a wallet w comprised of a single credential c_1, $w^{\text{single}}(c_1) = c_1$ we have built FAR vs FRR curves for Uniform and Quadratic probabilistic distribution functions and defined each one of the states as follows:

$$P_{\text{loss}} = \text{FRR} \cdot (1 - \text{FAR})$$
$$P_{\text{leak}} = \text{FAR} \cdot (1 - \text{FRR})$$
$$P_{\text{theft}} = \text{FRR} \cdot \text{FAR}$$
$$P_{\text{safe}} = 1 - \text{P}_{\text{loss}} - \text{P}_{\text{leak}} - \text{P}_{\text{theft}}$$

The probability for success of the wallet w, is defined as the probability of users successfully defend their wallets. In a single credential wallet, this is simply the probability that the key is neither theft, loss or leaked. i.e.,

$$P_{\text{success}}(w^{\text{single}}) = P_{\text{safe}} = 1 - \text{FAR} - \text{FRR} + \text{FAR} \cdot \text{FRR} \tag{1}$$

Until now, it has been common practice to select the operating point at the Equal Error Rate (EER), where FAR = FRR following Sarkar's claim: "The operating point is that point where the FAR is equal to the FRR and known as Equal Error Rate (EER)" [22]. Our findings indicate that a different operating point can yield a higher P_{safe}, thereby enhancing the system's overall security. Furthermore, We found that the operating point depends on the variance of the PDFs. We will further explore this in the following sections for different distributions.

4.1 Uniformly Distributed Credential

As a first-order approximation, we model the biometric credential using a uniform distribution. We defined two probability distribution functions for the user and the attacker, P_U and P_A respectively.

$$P_U(s) = \frac{1}{u_2 - u_1} \quad \text{for } u_1 \leq s \leq u_2$$
$$P_A(s) = \frac{1}{a_2 - a_1} \quad \text{for } a_1 \leq s \leq a_2$$

Where u_1, u_2 are the user PDFs' bounds and a_1, a_2 are the attacker PDFs' bounds.

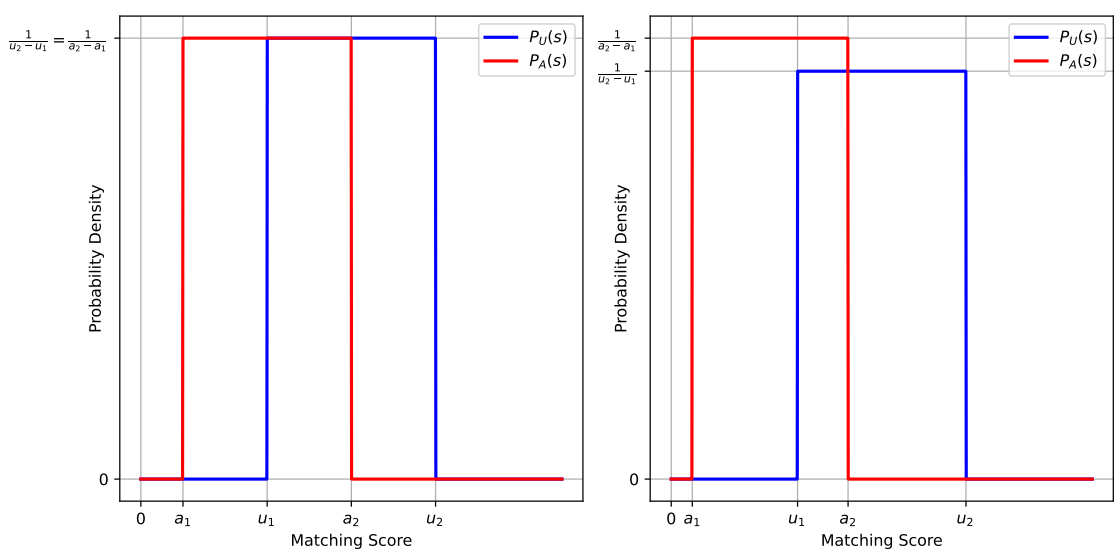


Figure 1: Symmetric and Asymmetric Uniform PDFs

We assume that $a_1 < u_1$ and $a_2 < u_2$ as the x axis represents the matching score s and higher values of s corresponds to the user D_1 as mentioned in the model details. Furthemore, we assume there is an overlaping region between the distributions; otherwise the scenario reduces to a trivial case. We also assume that the user and attacker PDFs are do not fully contain one another, as such cases would render them non-separable. Figure 1a illustrates the case where the user and attacker PDF's are symmetric, while Figure 1b illustrates asymmetric PDFs both as function of the matching score s. The probability of success for a single credential wallet is defined as:

$$P_{\text{success}}(w^{\text{single}}) = P_{\text{safe}} = 1 - \text{P}_{\text{loss}} - \text{P}_{\text{leak}} - \text{P}_{\text{theft}} = 1 - \text{FRR} - \text{FAR} + \text{FRR} \cdot \text{FAR}$$

Let T denote the threshold. Then the False Acceptance Rate (FAR) and False Rejection Rate (FRR) are defined as:

$$\text{FAR}(T) = \int_T^{a_2} \frac{1}{a_2 - a_1} ds = \begin{cases} 1 & \text{if } T \leq a_1 \\ \frac{a_2 - T}{a_2 - a_1} & \text{if } a_1 < T \leq a_2 \\ 0 & \text{if } T > a_2 \end{cases}$$

$$\text{FRR}(T) = \int_{u_1}^{T} \frac{1}{u_2 - u_1} ds = \begin{cases} 0 & \text{if } T \leq u_1 \\ \frac{T - u_1}{u_2 - u_1} & \text{if } u_1 < T \leq u_2 \\ 1 & \text{if } T > u_2 \end{cases}$$

We defined the probability of success as a function of the threshold T by setting the $FAR(T)$ $FRR(T)$ into the equation and received:

$$P_{\text{success}}(T) = \begin{cases} 0 & \text{if } T \leq a_1 \\ 1 - \frac{a_2 - T}{a_2 - a_1} & \text{if } a_1 < T \leq u_1 \\ 1 - \frac{T - u_1}{u_2 - u_1} - \frac{a_2 - T}{a_2 - a_1} + \frac{T - u_1}{u_2 - u_1} \cdot \frac{a_2 - T}{a_2 - a_1} & \text{if } u_1 < T \leq a_2 \\ 1 - \frac{T - u_1}{u_2 - u_1} & \text{if } a_2 < T \leq u_2 \\ 0 & \text{if } T > u_2 \end{cases}$$

$P_{\text{success}}(T)$ is a continuous function in $[a_1, u_2]$,differentiable in (a_1, u_2) and maintains $P_{\text{success}}(a_1) = P_{\text{success}}(u_2)$ thus by Rolle's Theorem, there exists at least one point in (a_1, u_2) such that:

$$\frac{d}{dT} P_{\text{success}}(T) = 0.$$

The horizontal tangent point is:

$$T = \frac{a_1 + u_2}{2} \quad \text{for } u_1 \leq T \leq a_2$$

$$\frac{d}{dt} P_{\text{success}}(T) > 0, \text{ For every } T \in (u_1, \frac{a_1 + u_2}{2})$$

$$\frac{d}{dt} P_{\text{success}}(T) < 0, \text{ For every } T \in (\frac{a_1 + u_2}{2}, a_2)$$

Thus, by the First Derivative Test the maximum point is at $T = \frac{a_1 + u_2}{2}$.

$$P_{\text{success}}(T_{\text{opt}}) = \frac{(a_1 - u_2)^2}{4(a_1 - a_2)(u_1 - u_2)}$$

We calculated the EER point and showed it is not equal to the optimal point:

$$T_{\text{EER}} = \frac{a_1 \cdot u_1 - a_2 \cdot u_2}{a_1 - a_2 + u_1 - u_2} \neq \frac{a_1 + u_2}{2}$$

The maximum point represnents the optimal threshold of the wallet. By selecting the threshold value, we influnce the wallet's security sensetivity by adjusting the probability associated with each state in S. The optimal point indicates that increasing the offset between the two uniform distributions is likely to improve the wallet's success. This optimal point suggest an asymmetry between the user's and the attacker's distribution intervals. Figure 2 illustrates the probability of success for a single credential wallet depends on the threshold T.

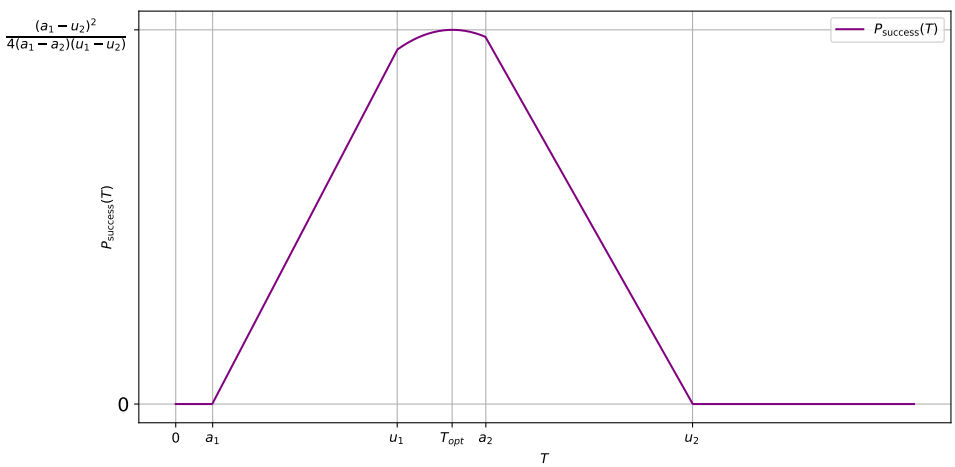


Figure 2: Probability of success for a single credential wallet asymmetrically distributed

We can define the FAR and FRR at the optimal point as follows:

$$\text{FAR}(T_{\text{opt}}) = \frac{2a_2 - a_1 - u_2}{2(a_2 - a_1)}$$

$$\text{FRR}(T_{\text{opt}}) = \frac{a_1 + u_2 - 2u_1}{2(u_2 - u_1)}$$

Figure 3 illustrated the FAR vs FRR curve.

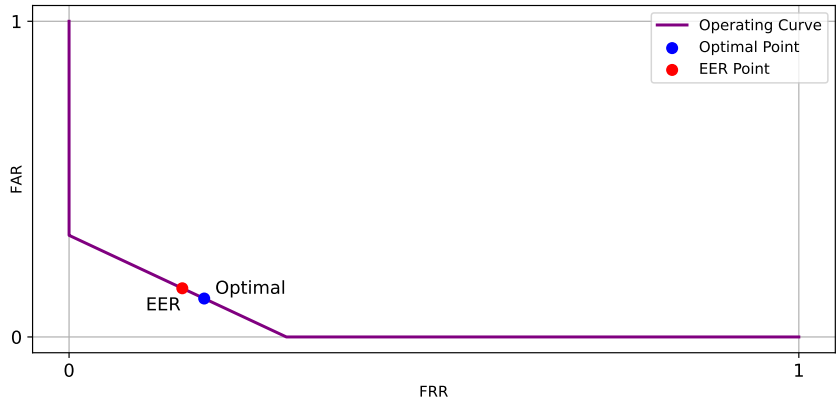


Figure 3: FAR vs FRR curve

4.2 Quadratic Distributed Credential

Our goal is to model credentials whose distributions closely resemble those observed in real measurement data. To this end, we analyze a quadratic probability distribution function, which provides a second order approximation of the biometric credential that can also be seen in real fingerprint measurement data (§7). This choice is also consistent with the findings of Slobodan et al. [21] on Eigenfinger and Eigenpalm feature distributions. Accordingly, we define two probability density functions (PDFs) for the user and the attacker, denoted by P_U and P_A, respectively.

$$P_U(s) = \begin{cases} a \cdot s^2 + b \cdot s + c & \alpha_1 \leq s \leq \alpha_2 \\ 0 & s > \alpha_2 \vee s < \alpha_1 \end{cases}$$

$$P_A(s) = \begin{cases} d \cdot s^2 + e \cdot s + f & \beta_1 \leq s \leq \beta_2 \\ 0 & s > \beta_2 \vee s < \beta_1 \end{cases}$$

Here, a, b, c and d, e, f denote the quadratic coefficients of the user and attacker distributions, respectively. The parameters α_1, α_2 are the roots of the user's PDF, while β_1, β_2 are the roots of the attacker's PDF. Similarly, to the analysis of the uniform distribution, the x axis represents the matching score, such that higher values of score will be corrspondant to the reference template stored in the database X_U. If $\beta_2 < \alpha_1$, the two distributions do not overlap, and the problem becomes trivial. Conversely, if $\alpha_1 > \beta_1$ and $\beta_2 < \alpha_2$, the attacker's distribution is fully contained within the user's, resulting in a non-separable scenario. We therefore focus on the intermediate and most meaningful configuration, where

$$\beta_1 \leq \alpha_1 \leq \beta_2 \leq \alpha_2,$$

as illustrated in Figure 4.

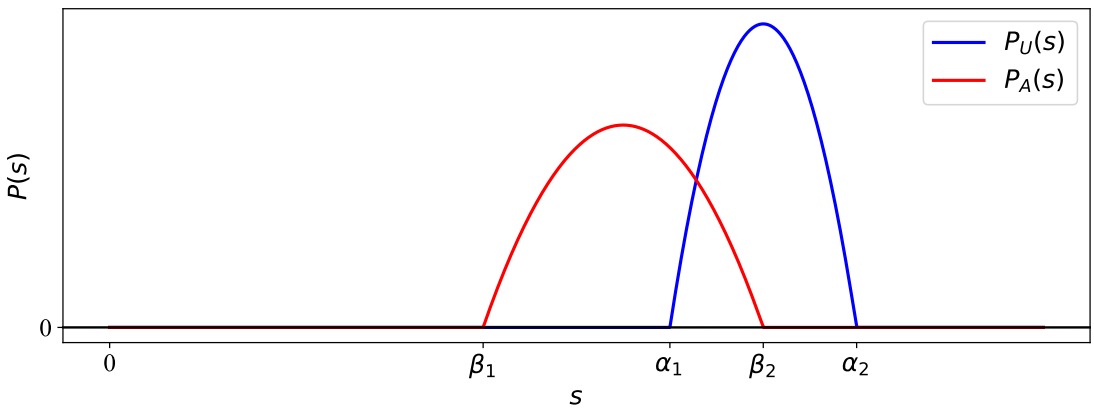


Figure 4: Quadratic PDFs

To reduce the degrees of freedom of the quadratic functions while maintaining a symbolic analysis, we express the coefficients directly in terms of their roots. Under the additional assumption that the integral of each PDF over its roots interval equals 1, the coefficients become fully determined. Accordingly, we define the coefficients as follows.

$$\alpha_1 \cdot \alpha_2 = \frac{c}{a} \rightarrow a = \frac{c}{\alpha_1 \cdot \alpha_2}$$

$$\alpha_1 + \alpha_2 = -\frac{b}{a} \rightarrow b = -\frac{c \cdot (\alpha_1 + \alpha_2)}{\alpha_1 \cdot \alpha_2}$$

$$\beta_1 \cdot \beta_2 = \frac{f}{d} \rightarrow d = \frac{f}{\beta_1 \cdot \beta_2}$$

$$\beta_1 + \beta_2 = -\frac{e}{d} \rightarrow e = -\frac{f \cdot (\beta_1 + \beta_2)}{\beta_1 \cdot \beta_2}$$

$$\int_{\alpha_1}^{\alpha_2} (a \cdot x^2 + b \cdot x + c) dx = 1 \Rightarrow c = \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3}$$

$$\int_{\beta_1}^{\beta_2} (d \cdot x^2 + e \cdot x + f) dx = 1 \Rightarrow f = \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3}$$

Therefore, we can write the quadratic PDFs as:

$$P_U(s) = \begin{cases} \frac{6}{(\alpha_1 - \alpha_2)^2} \cdot s^2 - \frac{6 \cdot (\alpha_1 + \alpha_2)}{(\alpha_1 - \alpha_2)^3} \cdot x + \frac{6 \cdot \alpha_1 \cdot \alpha_2}{(\alpha_1 - \alpha_2)^3} & \alpha_1 \leq x \leq \alpha_2 \\ 0 & x > \alpha_2 \vee x < \alpha_1 \end{cases}$$

$$P_A(s) = \begin{cases} \frac{6}{(\beta_1 - \beta_2)^2} \cdot x^2 - \frac{6 \cdot (\beta_1 + \beta_2)}{(\beta_1 - \beta_2)^3} \cdot x + \frac{6 \cdot \beta_1 \cdot \beta_2}{(\beta_1 - \beta_2)^3} & \beta_1 \leq x \leq \beta_2 \\ 0 & x > \beta_2 \vee x < \beta_1 \end{cases}$$

We will find FAR and FRR as a function of the threshold T and define the probability of success based on the single credential wallet as shown in (§4.1):

$$\text{FAR}(T) = \int_T^{\beta_2} P_A(s) ds = \begin{cases} 0 & T \leq \beta_1 \\ \frac{6\left(\frac{\beta_2^3}{3} - \frac{T^3}{3}\right)}{(\beta_1 - \beta_2)^3} - \frac{6\left(\frac{\beta_2^2}{2} - \frac{T^2}{2}\right)(\beta_1 + \beta_2)}{(\beta_1 - \beta_2)^3} + \frac{6\beta_1\beta_2(\beta_2 - T)}{(\beta_1 - \beta_2)^3} & \beta_1 < T \leq \beta_2 \\ 1 & T > \beta_2 \end{cases}$$

$$\mathrm{FRR}(T) = \int_{\alpha_1}^{T} P_U(s)ds = \begin{cases} 1 & T \le \alpha_1 \\ \frac{6\left(\frac{T^3}{3} - \frac{\alpha_1^3}{3}\right)}{(\alpha_1-\alpha_2)^3} - \frac{6\left(\frac{T^2}{2} - \frac{\alpha_1^2}{2}\right)(\alpha_1+\alpha_2)}{(\alpha_1-\alpha_2)^3} + \frac{6\alpha_1\alpha_2(T-\alpha_1)}{(\alpha_1-\alpha_2)^3} & \alpha_1 < T \le \alpha_2 \\ 0 & T > \alpha_2 \end{cases}$$

$$P_{\text{success}} = \begin{cases} 0 & T < \beta_1 \vee T > \alpha_2 \\ \frac{(T-\beta_1)^2(2T+\beta_1-3\beta_2)}{(\beta_1-\beta_2)^3} & \beta_1 \le T \le \alpha_1 \\ -\frac{(T-\alpha_2)^2(T-\beta_1)^2(2T-3\alpha_1+\alpha_2)(2T+\beta_1-3\beta_2)}{(\alpha_1-\alpha_2)^3(\beta_1-\beta_2)^3} & \alpha_1 \le T \le \beta_2 \\ -\frac{(T-\alpha_2)^2(2T-3\alpha_1+\alpha_2)}{(\alpha_1-\alpha_2)^3} & \beta_2 \le T \le \alpha_2 \end{cases}$$

We differentiate $P_{\text{success}}(T)$ and compute its critical points in each interval. By comparing the stationary values obtained in all valid regions, we identify the global maximum, which corresponds to the optimal threshold T_{opt}.

Before writing the piecewise derivative, we define the polynomial coefficients:

$$q = 4,$$

$$m = -(5\alpha_1 + \alpha_2 + \beta_1 + 5\beta_2),$$

$$n = \alpha_1(3\alpha_2 + \beta_1 + 6\beta_2) - \alpha_2^2 + \alpha_2\beta_2 - \beta_1^2 + 3\beta_1\beta_2,$$

$$\ell = \alpha_1\big(\beta_1^2 - 3\beta_2(\alpha_2 + \beta_1)\big) + \alpha_2^2\beta_2.$$

With these coefficients, the derivative of $P_{\text{success}}(T)$ is given by the following piecewise expression:

$$\frac{dP_{\text{success}}}{dT} = \begin{cases} 0, & T < \beta_1 \ \vee \ T > \alpha_2, \\ \dfrac{6(T-\beta_1)(T-\beta_2)}{(\beta_1-\beta_2)^3}, & \beta_1 \le T \le \alpha_1, \\ -\dfrac{6(T-\alpha_2)(T-\beta_1)\big(qT^3 + mT^2 + nT + \ell\big)}{(\alpha_1-\alpha_2)^3(\beta_1-\beta_2)^3}, & \alpha_1 \le T \le \beta_2, \\ -\dfrac{6(T-\alpha_1)(T-\alpha_2)}{(\alpha_1-\alpha_2)^3}, & \beta_2 \le T \le \alpha_2. \end{cases}$$

In the second interval, the derivative has a single critical point at $T = \beta_1$. Since the derivative is strictly positive for all $T > \beta_1$, this point corresponds to a local minimum rather than a maximum. Similarly, in the fourth interval, the only critical point occurs at $T = \alpha_2$, and the derivative is strictly negative for all $T < \alpha_2$. Therefore, this interval also contains no maximum either.

We therefore turn to the third interval. In this region, the derivative is the product of a cubic polynomial and two linear factors. The linear factors vanish at $T = \beta_1$ and $T = \alpha_2$, both of which lie outside the third interval. Consequently, the existence of critical points within this interval depends solely on the cubic term. Solving this cubic ,reveals exactly one real root with the remaining two roots being complex. The closed form of the cubic solution can not be simplified further more than a complex expression. Thus, we denote the solution as T_{cubic}.

By examining the sign of the derivative on both sides of the critical point T_{cubic}, we verify that it constitutes a global maximum of $P_{\text{safe}}(T)$. We denote this maximizing point by T_{opt}.

To demonstrate that this optimal threshold differs from the Equal Error Rate (EER) threshold, we computed the EER point separately as follows:

$$T_{\text{EER}} = \frac{\alpha_1\beta_1 - \alpha_2\beta_2}{\alpha_1 - \alpha_2 + \beta_1 - \beta_2} \neq T_{\text{opt}}$$

We illustrate several user and attacker distributions and present their corresponding probability of success curves, expressed as functions of the threshold T in Figure 5 and Figure 6 respectively.

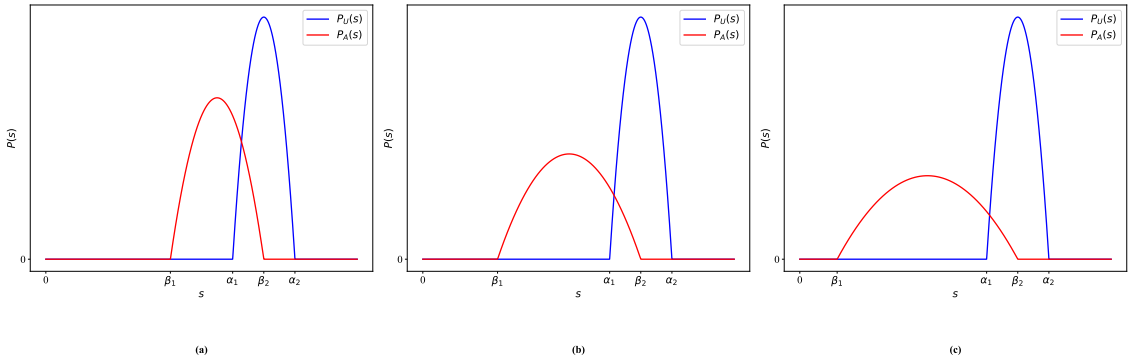


Figure 5: User and Attacker Quadratically Distributed

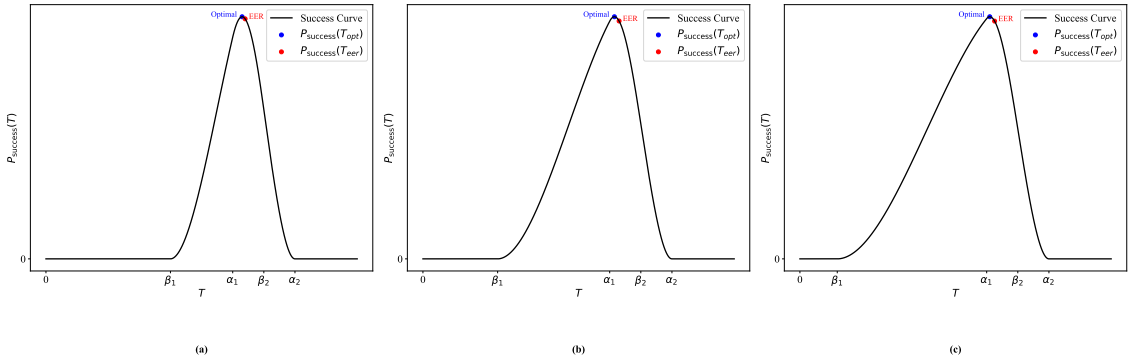


Figure 6: Probability of success for a single credential wallet quadratically distributed

Moreover, we illustrate the FAR–FRR curves for each pair of user and attacker distributions, and mark both the optimal point and the EER point, as shown in Figure 7.

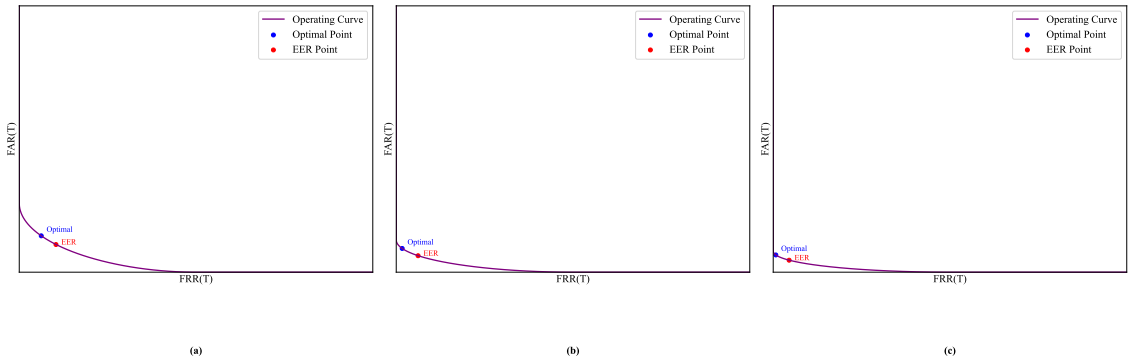


Figure 7: FAR vs FRR curve quadratically distributed

To quantify the effect of the variance on the deviation between the optimal operating point T_{opt} and the equal error rate (EER) point T_{EER}, we numerically computed this gap as a function of the attacker's variance. Decreasing the variance of the attacker achieved by reducing the distance between its roots, corresponds to reduced separability between the two distributions. The simulations reveal that as the variance increases, the gap $|T_{\text{opt}} - T_{\text{EER}}|$ grows accordingly, until the operating threshold moves into a regime where it effectively minimizes only one of the error rates, either FAR or FRR, driving the other to zero. This phenomenon is shown in Figure 8. These findings indicate that higher attacker variance enlarges the potential security advantage gained by operating at T_{opt} rather than at the EER point.

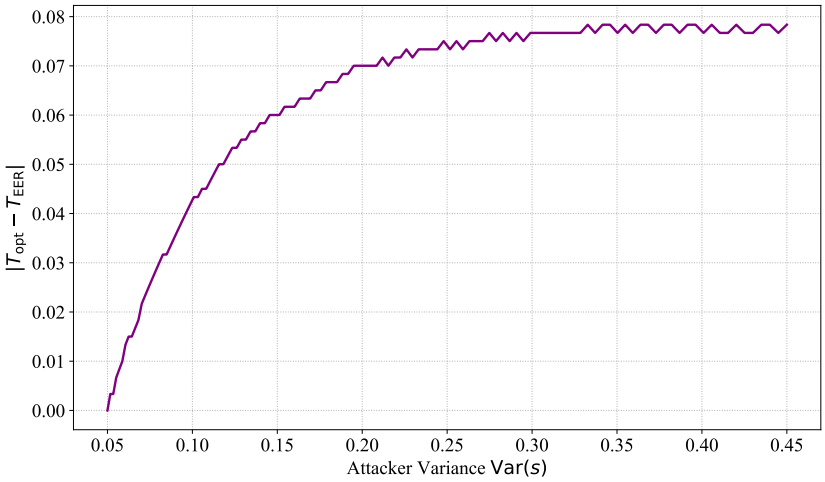


Figure 8: Gap between optimal point and EER point vs Variance of the attacker quadratically distributed

5 Two Credentials

For a wallet w comprised of two independently and identically distributed credentials, derived from either the same or different PDF's, we defined the integrated system success as the probability of users successfully defend their wallets under two scenarios:

a. **AND** Require both credentials to be satisfied. For success, we need both credentials to be either in safe state, or one in safe state and the other in leak state. If one of the credentials is either loss ot theft, the owner cannot satisfy the wallet and thus the wallet fails.

$$P_{\text{Success}}(w^{\text{AND}}) = \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{safe}} + \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{leak}} + \text{P}^2_{\text{safe}} \cdot \text{P}^1_{\text{leak}} \tag{2}$$

b. **OR** Require at least one credential to be satisfied. For success, we need both credential to be either in safe state, or one in safe state and the other in loss state. If one of the credentials is either leak ot theft, the attacker can satisfy the wallet and thus the wallet fails.

$$P_{\text{Success}}(w^{\text{OR}}) = \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{safe}} + \text{P}^1_{\text{safe}} \cdot \text{P}^2_{\text{loss}} + \text{P}^2_{\text{safe}} \cdot \text{P}^1_{\text{loss}} \tag{3}$$

For each wallet type, we analyzed the case of two independently and identically distributed credentials in 2 cases:

a. A mechanism comprised of 2 biometric credentials.

b. A mechanism comprised of 1 discrete credential and 1 biometric credential.

We derived the optimal operating point for both **AND** and **OR** wallets by maximizing the respective success probabilities with respect to the thresholds of each credential.

5.1 2 Biometric Credentials

To analyze the case of a wallet w comprised of two biometric credentials, we conduct a numerical study over several probability distribution functions. For each distribution pair, we derive the optimal operating point for both **AND** and **OR** wallets by maximizing the respective success probabilities with respect to the thresholds of each credential, using a LBFGS optimization algorithm. We additionally compare the optimal operating point to the EER point for each distribution type, thereby characterizing the conditions under which the **AND** or **OR** wallet yields superior performance. We started by analyzing the case of Gaussian distributions. We defined the user and attacker distributions as $P_U(s) = \mathcal{N}(\mu_u, \sigma_u^2)$ and $P_A(s) = \mathcal{N}(\mu_a, \sigma_a^2)$ respectively. We set for the first credential $\mu_u = 0.8$, $\sigma_u = 0.12$, $\mu_a = 0.3$ and $\sigma_a = 1$ and for the second credential $\mu_u = 1.0$, $\sigma_u = 0.2$, $\mu_a = 0.4$ and $\sigma_a = 0.18$ as shown in Figure 9.

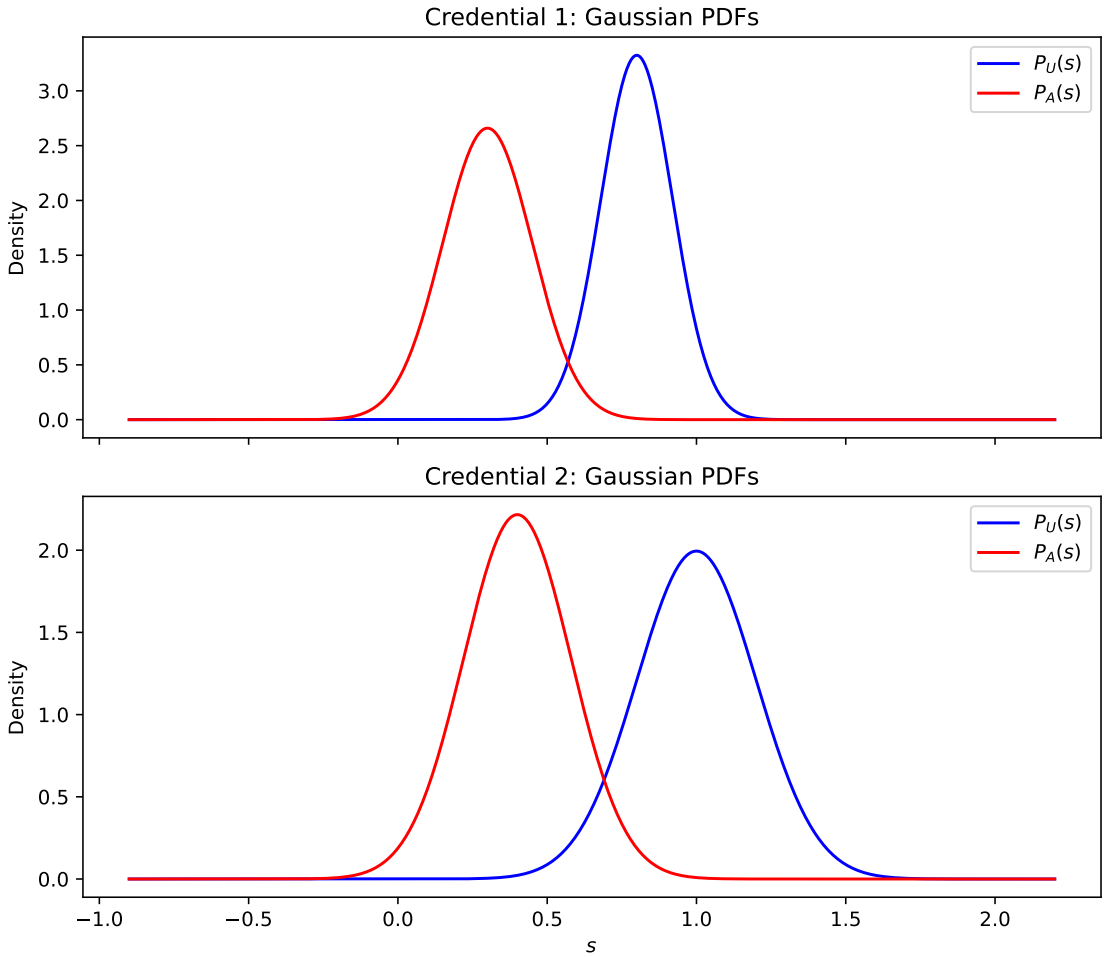


Figure 9: Gaussian distributed credentials

The optimal operating point for both wallet types was found to differ from the EER point, with the optimal point yielding a higher success probability than the EER point. Moreover, we found out that optimal point of both **AND** and **OR** wallets shifts from the individual optimum point of the credential. Hence, integrating the two credentials in a wallet improves performance over using them individually as shown in Figure 10.

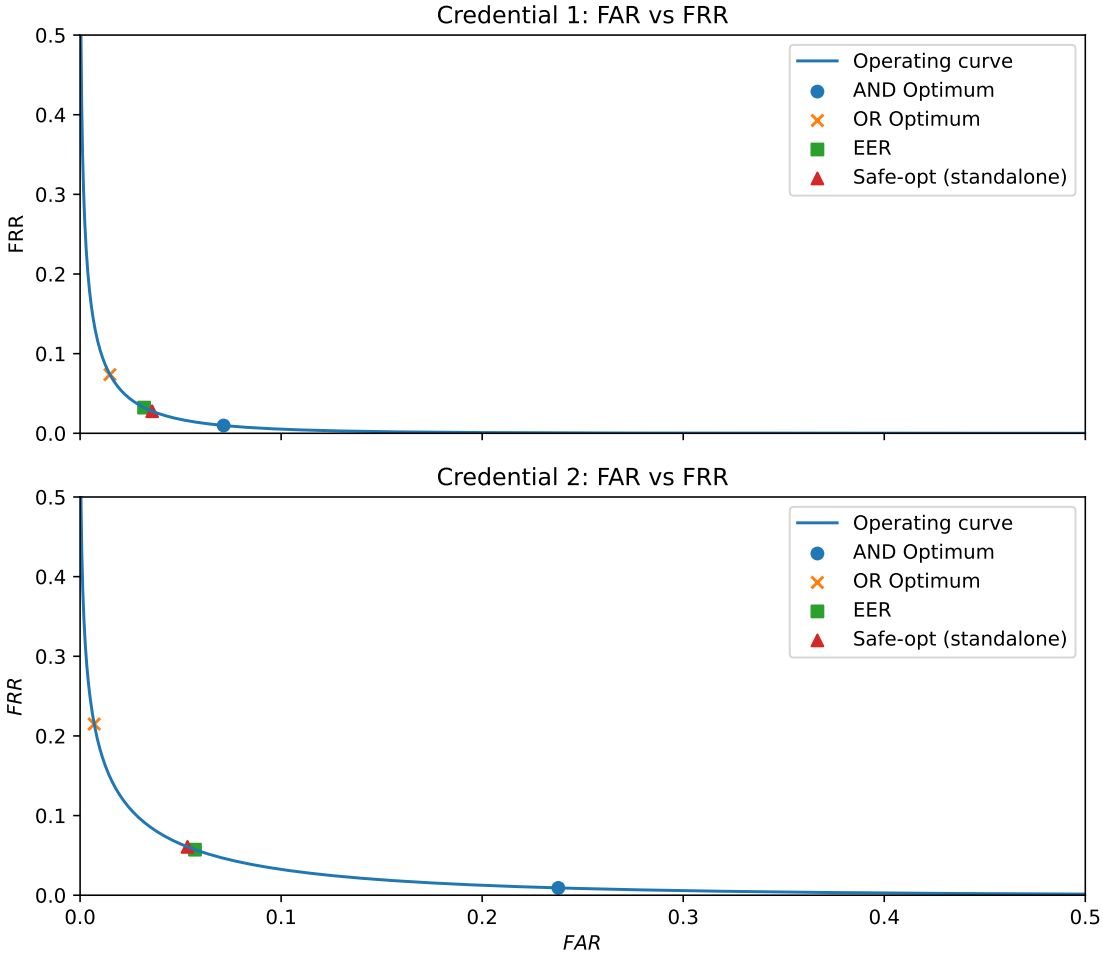


Figure 10: FAR vs FRR curves for the two Gaussian credentials

We illustrated the success probability of both **AND** and **OR** wallets. We fixed a credential to its optimum found out using the LBFGS algorithm and showed how the success probability varies as a function of the thresholds of the other credential. The results are shown in Figure 11. In conclusion, all the results are summarized in Table 1.

Distribution	$P_{\text{AND}}(T_{\text{EER}})$	$P_{\text{OR}}(T_{\text{EER}})$	$P_{\text{AND}}(T_1^{\text{safe}})$	$P_{\text{OR}}(T_2^{\text{safe}})$	$\boldsymbol{P_{\text{AND}}(T_{\text{opt}})}$	$\boldsymbol{P_{\text{OR}}(T_{\text{opt}})}$
Gaussian	0.910	0.911	0.911	0.911	0.964	0.962
Uniform	0.955	0.955	0.916	0.999	1.000	1.000
Parabolic	0.920	0.921	0.919	0.922	0.987	0.985

Table 1: Wallet success under Gaussian, Uniform and Parabolic score distributions at three reference operating points: EER-pair $(T_1^{\text{eer}}, T_2^{\text{eer}})$, individual optimum pair $(T_1^{\text{safe}}, T_2^{\text{safe}})$, and joint optima (AND at $(T_1^*, T_2^*)_{\text{AND}}$ and OR at $(T_1^*, T_2^*)_{\text{OR}}$)

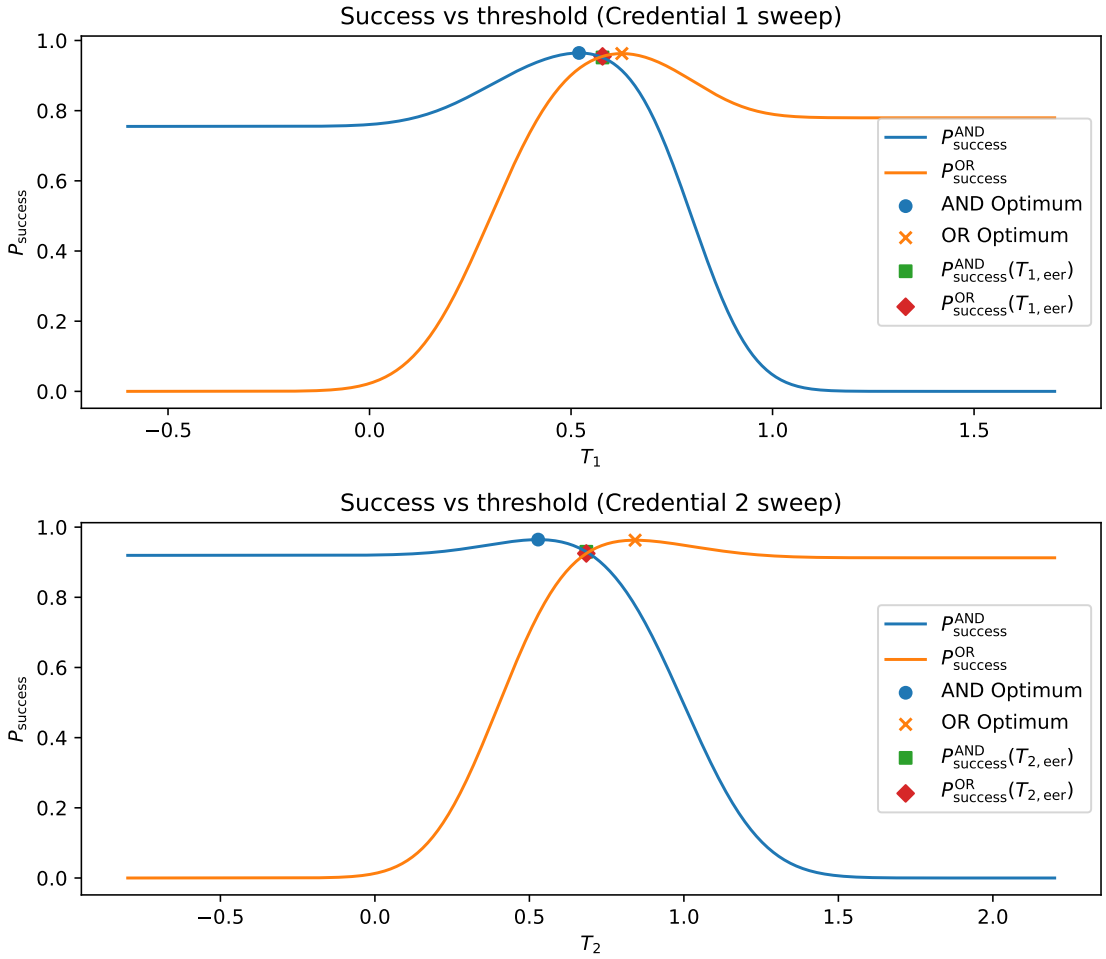


Figure 11: Success probability functions for the two Gaussian credentials

From Table 1, we observe that the optimal operating point yields a higher success probability than the EER point by between 5% to 7%.

5.2 1 Binary and 1 Biometric Credential

We started by analyzing the case of an **AND** wallet comprised of 1 binary credential and 1 biometric credential. Denote s_i the state of credential $i \in (Binary, Biometric)$, we defined the binary credential with a constant value of each one of the probabilities' states as: $P^1_{safe}, P^1_{loss}, P^1_{leak}, P^1_{theft}$ corrsponding respectively to each one of the states: safe, loss, leak and theft. The biometric credential is defined by its PDFs for the user and the attacker, P_U and P_A respectively. We firstly modeled the first order approximation of the biometric credential using uniform distribution between the bounds of u_1,u_2 corrsponding to the user distribution and a_1,a_2 corresponds to the attacker, as seen at (§4.1). We derived the probability of the biometric credential as follows:

$$P^2_{safe} = 1 - \mathrm{FAR}(T) - \mathrm{FRR}(T) + \mathrm{FAR}(T) \cdot \mathrm{FRR}(T),$$
$$P^2_{leak} = \mathrm{FAR}(T)\big(1 - \mathrm{FRR}(T)\big).$$

To analyze the probability of success for the **AND** wallet, we set the above probabilities into Equation 2. Thus, the probability of success is a function of the threshold T and divided into 5 intervals.

$$P_{\text{success}}(T) = \begin{cases} P^1_{\text{safe}}, & T \le a_1, \\ (P^1_{\text{safe}} + P^1_{\text{leak}}) - P^1_{\text{leak}} \dfrac{a_2 - T}{a_2 - a_1}, & a_1 < T \le u_1, \\ \dfrac{u_2 - T}{u_2 - u_1} \left[(P^1_{\text{safe}} + P^1_{\text{leak}}) - P^1_{\text{leak}} \dfrac{a_2 - T}{a_2 - a_1} \right], & u_1 < T \le a_2, \\ \dfrac{u_2 - T}{u_2 - u_1} (P^1_{\text{safe}} + P^1_{\text{leak}}), & a_2 < T \le u_2, \\ 0, & T > u_2 \end{cases}$$

We analyze the success probability $P_{\text{success}}(T)$ by partitioning the threshold domain into five intervals, each corresponding to a distinct relative positioning of the threshold with respect to the attacker and user distributions.

The first interval, $T \le a_1$, corresponds to the case in which the threshold is lower than the minimum value of the attacker distribution. In this regime, the FAR is equal to 1, while the FRR is equal to 0. Consequently, the safe probability of the biometric credential is zero, and the success probability converges to P^1_{safe}. Since P^1_{safe} is constant in this interval, no interior maximum exists.

The second interval, $a_1 < T \le u_1$, corresponds to the case in which the threshold lies between the minimum value of the attacker distribution and the minimum value of the user distribution. In this region, the FAR decreases linearly as a function of T, while the FRR remains equal to 0. As a result, the probabilities of the safe state and the leak state of the biometric credential are $1 - \mathrm{FAR}(T)$ and $\mathrm{FAR}(T)$, respectively. In this interval, $P_{\text{success}}(T)$ is a strictly increasing function, as its derivative is a constant positive value, and therefore the maximum is attained at the boundary point

$$T_{\text{opt2}} = u_1.$$

The third interval, $u_1 < T \le a_2$, corresponds to the case in which the threshold lies between the minimum value of the user distribution and the maximum value of the attacker distribution. In this region, both FAR and FRR vary linearly with T, resulting in a success probability that

is quadratic as a function of T. Differentiating $P_{\text{success}}(T)$ yields a single critical point inside the interval,

$$T_{\text{opt3}} = \frac{u_2 + a_2 - (a_2 - a_1)\,\dfrac{P^1_{\text{safe}} + P^1_{\text{leak}}}{P^1_{\text{leak}}}}{2}$$

Since this expression is valid only for the interval $u_1 < T \leq a_2$, the effective maximizer within this interval is obtained by projecting the unconstrained optimum onto the feasible domain:

$$T^{(\text{interval})}_{\text{opt},3} = \min(a_2,\ \max(u_1,\ T_{\text{opt},3}))\,.$$

In particular, if $T_{\text{opt},3} < u_1$, the maximum over this interval is attained at $T = u_1$, whereas if $T_{\text{opt},3} > a_2$, the maximum is attained at $T = a_2$. In all configurations consisting of one binary and one biometric credential, ingertating the wallet with the biometric credential monotonically enhances its security. Evaluating the success probability at the user threshold $T = u_1$ yields

$$P_{\text{success}}(u_1) = P^1_{\text{safe}} + P^1_{\text{leak}}(1 - \frac{a_2 - u_1}{a_2 - a_1}).$$

Since $u_1 > a_1$, it follows that

$$P_{\text{success}}(u_1) > P^1_{\text{safe}},$$

Demonstrating that the integration of the biometric credential strictly improves the security of the wallet compared to using the binary credential alone.

By examining the optimal point in this interval, we can deduce that as the variance of the attacker distribution increases, the optimal threshold T_{opt3} shifts toward the lower boundry of the interval, u_1. Moreover, for a binary credential with high safe values P^1_{safe}, and low leak values P^1_{leak}, the optimal threshold shifts toward the lower boundary as well. Thus, in such case, the success of the wallet becomes mainly dependent on the binary credential. Conversely, a binary credential with high leak values P^1_{leak} and low safe values P^1_{safe} causes the optimal threshold to shift toward the upper boundary of the interval, a_2. In this scenario, the success of the wallet relies more heavily on the biometric credential.

The fourth interval, $a_2 < T \leq u_2$, corresponds to the case in which the threshold exceeds the maximum value of the attacker distribution. In this regime, the FAR is equal to 0, while the FRR increases linearly with T. Consequently, $P_{\text{success}}(T)$ is a strictly decreasing function in this interval, and the maximum is attained at the boundary point

$$T_{\text{opt4}} = a_2.$$

Finally, the fifth interval, $T > u_2$, corresponds to the case in which the threshold exceeds the maximum value of the user and the attacker distributions, leaving the success probability equal to 0.

To determine the global optimal operating point, we compare all stationary points and boundary maxima obtained in the valid intervals. The optimal threshold is achieved in the third interval, where the attacker and user distributions overlap and the success function attains an interior maximum. Substituting $T_{\text{opt}} = T_{\text{opt3}}$ into $P_{\text{success}}(T)$ yields

$$P_{\text{success}}(T_{\text{opt}}) = \frac{\left(-a_1\left(P^1_{\text{safe}} + P^1_{\text{leak}}\right) + a_2 P^1_{\text{safe}} + P^1_{\text{leak}}\, u_2\right)^2}{4P^1_{\text{leak}}(a_2 - a_1)(u_2 - u_1)}.$$

We have done similar analysis for the **OR** wallet. We defined the credentials' probabilities of the wallet as follows:

$$\begin{aligned}
P_{\text{safe1}} &= 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T)\cdot\text{FRR}(T),\\
P_{\text{safe2}} &= k_1,\\
P_{\text{loss1}} &= \text{FRR}(T)\left(1 - \text{FAR}(T)\right),\\
P_{\text{loss2}} &= k_2.
\end{aligned}$$

Therefore, the probability of success for the **OR** wallet is defined as:

$$P_{\text{success}}(T) = \begin{cases} 0, & T \leq a_1, \\ \dfrac{T - a_1}{a_2 - a_1}\,(k_1 + k_2), & a_1 < T \leq u_1, \\ \dfrac{T - a_1}{a_2 - a_1}\left[k_1 + k_2 - k_2\,\dfrac{T - u_1}{u_2 - u_1}\right], & u_1 < T \leq a_2, \\ k_1 + k_2 - k_2\,\dfrac{T - u_1}{u_2 - u_1}, & a_2 < T \leq u_2, \\ k_1, & T > u_2. \end{cases}$$

By differentiating every interval and identifying the critical points, we found that the optimal threshold that leads to the maximum probability of success lies in the third interval where $u_1 < T \leq a_2$ and is given by:

$$T_{\text{opt}} = \frac{a_1 k_2 + u_2(k_1 + k_2) - k_1 u_1}{2k_2}$$

And the maximum probability of success at this point is:

$$P_{\text{success}}(T_{\text{opt}}) = \frac{\big(a_1 k_2 - u_2(k_1 + k_2) + k_1 u_1\big)^2}{4k_2(a_2 - a_1)(u_2 - u_1)}$$

For both wallet types, we observed that the optimal operating point T_{opt} differs from the equal error rate (EER) point T_{EER}. As the EER depends on the PDF's type solely, we calculated the EER point and showed each wallet's success in Figure 12 and Figure 13 respectively.

$$T_{\text{EER}} = \frac{a_1 u_1 - a_2 u_2}{a_1 - a_2 + u_1 - u_2}$$

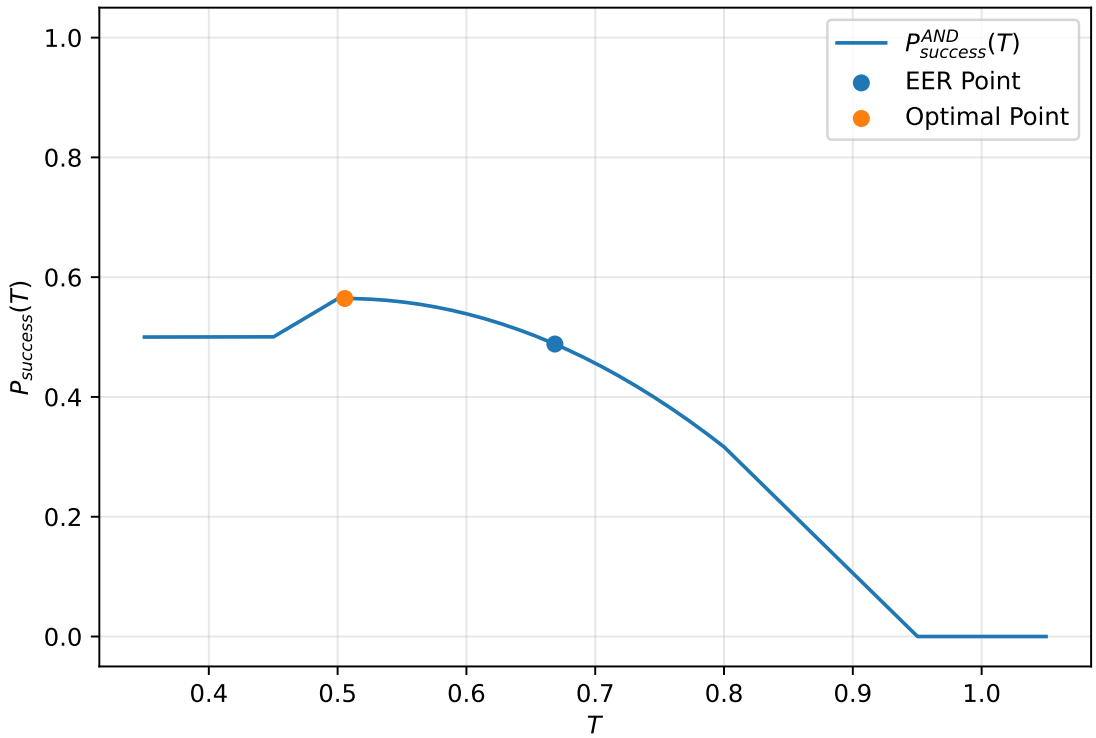


Figure 12: Success probability functions for one binary and one biometric credential in an AND wallet

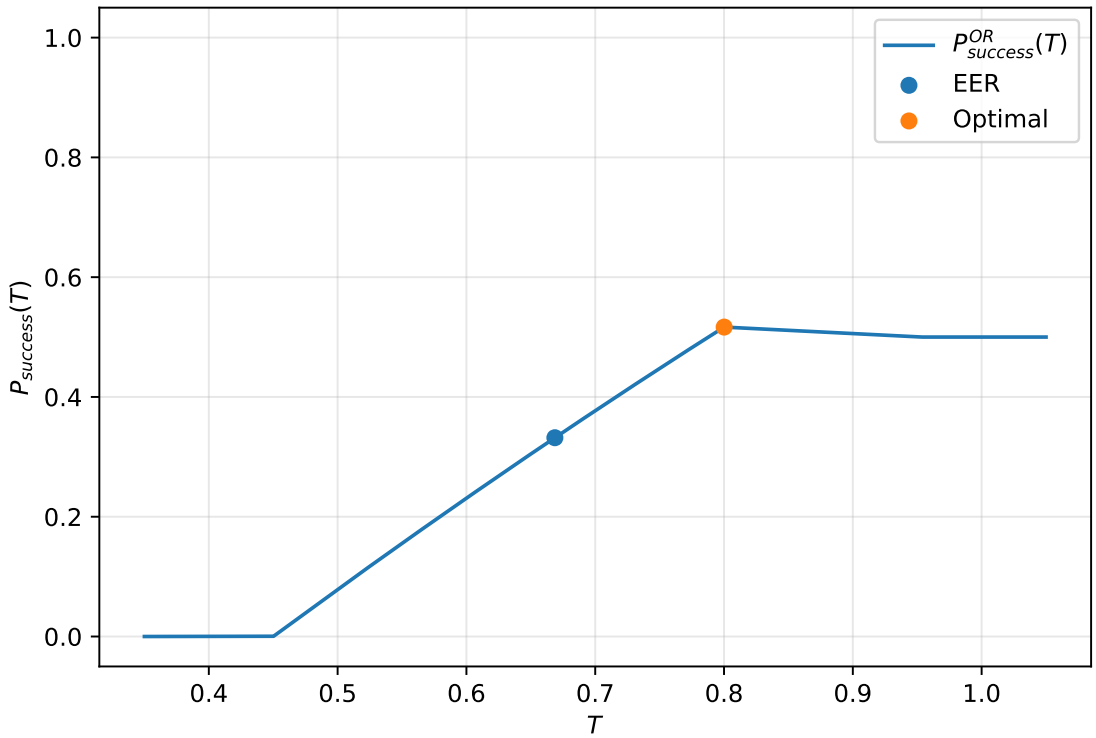


Figure 13: Success probability functions for one binary and one biometric credential in an OR wallet

6 Characterizing Security Improvement

We will now characterize the conditions under which integrating a biometric credential into a wallet composed of a single binary credential can enhance its security. First, we show that for any wallet whose binary credential characterized by non-zero probabilities P_{safe} and P_{theft} only, such integration cannot improve security (§6.1). Next, we present a necessary and sufficient condition establishing that, for any wallet composed of a binary credential with non zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss}, integrating a biometric credential improves security for both finite and infinite distributions (§6.2). Finally, we provide a more intuitive sufficient condition under which a biometric credential enhances the security of a wallet based on a single binary credential with non zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss} (§6.3).

6.1 No Security Gain in the Safe/Theft Case

Proposition 1. *For any wallet w composed of a single binary credential with non-zero probabilities P_{safe} and P_{theft} only, integrating the wallet with a biometric credential does not enhance its security.*

Proof. Consider a wallet w composed of a single binary credential with non-zero probabilities

P_{safe} and P_{theft} only. In this case, integrating the wallet with a biometric credential lead to success probability described by Equation 2 and Equation 3. Since $P_{\text{leak}} = P_{\text{loss}} = 0$, the success probability in both wallet cases reduces to:

$$P_{\text{success}}(T) = P^1_{\text{safe}} \cdot P^2_{\text{safe}}(T)$$

Since $P^2_{\text{safe}}(T) \leq 1$ for every value of T, it follows that $P_{\text{success}}(T) \leq P^1_{\text{safe}}$, and thus integrating a biometric credential does not enhance the security of the wallet. □

6.2 Sufficient And Necessary Condition for Integrating Biometric Credentials

First, we will define the notion of *asymptotic separability* of a biometric credential. Then, we will show that its a necessary and sufficient condition for enhancing the security for both finite and infinite distributions.

Definition 1. A biometric credential with user distribution function $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically separable* if

$$\inf_t \min\left\{\frac{\text{FAR}(t)}{1-\text{FRR}(t)}, \frac{\text{FRR}(t)}{1-\text{FAR}(t)}\right\} = 0,$$

where $\text{FAR}(t) = \int_t^\infty P_a(s)ds$ and $\text{FRR}(t) = \int_{-\infty}^t P_u(s)ds$.

Proposition 2. *For any wallet w composed of a single binary credential with non-zero probabilities P_{safe}, P_{leak} or P_{loss}, integrating the wallet with a biometric credential whose user and attacker distributions are asymptotically separable, enhances the security of the wallet.*

Proof. The security of a wallet composed of a single binary credential with non-zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss} is given by P^1_{safe}. A biometric credential is characterized by 4 probability states as function of T: $P^2_{\text{safe}}(T), P^2_{\text{loss}}(T), P^2_{\text{leak}}(T), P^2_{\text{theft}}(T)$. We can describe each one of these probabilities as a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$\begin{aligned}
P^2_{\text{safe}}(T) &= 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T)\text{FRR}(T),\\
P^2_{\text{loss}}(T) &= \text{FRR}(T)\big(1 - \text{FAR}(T)\big),\\
P^2_{\text{leak}}(T) &= \text{FAR}(T)\big(1 - \text{FRR}(T)\big),\\
P^2_{\text{theft}}(T) &= \text{FAR}(T)\text{FRR}(T).
\end{aligned}$$

The success probability of the integrated wallet comprised of 2 credentials is described by Equation 2 and Equation 3. Thus, by placing the biometric credential in each configuration we can express the success probability of the integrated wallet as a function of the operating threshold T:

$$P^{\text{AND}}_{\text{success}}(T) = \big(1 - \text{FRR}(T)\big)\Big(P^1_{\text{safe}} + P^1_{\text{leak}}\big(1 - \text{FAR}(T)\big)\Big). \tag{4}$$

$$P^{\text{OR}}_{\text{success}}(T) = \big(1 - \text{FAR}(T)\big)\Big(P^1_{\text{safe}} + P^1_{\text{loss}}\big(1 - \text{FRR}(T)\big)\Big). \tag{5}$$

We need to show that there exists an operating threshold T such that either $P^{\text{AND}}_{\text{success}}(T) > P^1_{\text{safe}}$ or $P^{\text{OR}}_{\text{success}}(T) > P^1_{\text{safe}}$.

By algebraic manipulation of Equation 4 we can demand $P^{\text{AND}}_{\text{success}}(T) > P^1_{\text{safe}}$ if and only if:

$$\frac{\text{FRR}(T)}{1 - \text{FAR}(T)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}} \cdot (1 - \text{FAR}(T))}$$

Since $P^1_{\text{leak}} \cdot (1 - \text{FAR}(T)) < P^1_{\text{leak}}$, a sufficient condition for the above inequality is:

$$\frac{\text{FRR}(T)}{1 - \text{FAR}(T)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}}} \tag{6}$$

Similarly, we can deduce that $P^{\text{OR}}_{\text{success}}(T) > P^1_{\text{safe}}$ if and only if:

$$\frac{\text{FAR}(T)}{1 - \text{FRR}(T)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}} \cdot (1 - \text{FRR}(T))}$$

Since $P^1_{\text{loss}} \cdot (1 - \text{FRR}(T)) < P^1_{\text{loss}}$, a sufficient condition for the above inequality is:

$$\frac{\text{FAR}(T)}{1 - \text{FRR}(T)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}}} \tag{7}$$

By the definition of asymptotic separability, there exists an operating threshold T such that:

$$\inf_t \min\left\{ \frac{\text{FAR}(t)}{1 - \text{FRR}(t)}, \frac{\text{FRR}(t)}{1 - \text{FAR}(t)} \right\} = 0.$$

Equivalently, $\forall \varepsilon > 0, \exists T$ such that either $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} < \varepsilon$ or $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} < \varepsilon$. Thus, we can choose T such that ε satisfies either Equation 6 or Equation 7. More precisely, if

$$\inf_T \frac{\text{FAR}(T)}{1 - \text{FRR}(T)} = 0,$$

then the OR wallet composition is preferable, whereas if

$$\inf_T \frac{\text{FRR}(T)}{1 - \text{FAR}(T)} = 0,$$

then the AND wallet composition is preferable. Therefore, integrating an asymptotically separable biometric credential enhances the security of the wallet.

□

Proposition 3. *Suppose there $\exists T$ such that the integrated wallet w composed of biometric and binary credential with non zero probabilities P_{safe}, P_{leak} or P_{loss} achieves a strictly higher success probability than the binary-only wallet, i.e.,*

$$P^{integrated}_{success}(T) > P^{binary}_{success}.$$

Then necessarily, the biometric credential is asymptotically separable.

Proof. We prove by contradiction. Assume for the sake of contradiction, that the biometric credential is not asymptotically separable. Then $\exists \varepsilon > 0$ such that $\forall T$ either $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} \geq \varepsilon$ or $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} \geq \varepsilon$. Since it's given that the integrated wallet w achieves a strictly higher success probability than the binary-only wallet, then there exists an operating threshold T such that either $P^{\text{AND}}_{\text{success}}(T) > P^1_{\text{safe}}$ or $P^{\text{OR}}_{\text{success}}(T) > P^1_{\text{safe}}$. Thus, either Equation 6 or Equation 7 holds for every value of $P^1_{\text{safe}}, P^1_{\text{leak}}, P^1_{\text{loss}}$ in the range of $(0, 1)$. In particular, we can choose $P^1_{\text{safe}}, P^1_{\text{leak}}, P^1_{\text{loss}}$ such that $\forall \varepsilon > 0 : \frac{P^1_{\text{leak}}}{P^1_{\text{safe}}+P^1_{\text{leak}}} < \varepsilon$ and $\frac{P^1_{\text{loss}}}{P^1_{\text{safe}}+P^1_{\text{loss}}} < \varepsilon$. Thus, we can deduce that both $\frac{\text{FRR}(T)}{1-\text{FAR}(T)} \geq \frac{P^1_{\text{leak}}}{P^1_{\text{safe}}+P^1_{\text{leak}}}$ and $\frac{\text{FAR}(T)}{1-\text{FRR}(T)} \geq \frac{P^1_{\text{loss}}}{P^1_{\text{safe}}+P^1_{\text{loss}}}$, which contradicts the assumption that the integrated wallet achieves a strictly higher success probability than the binary-only wallet. Therefore, the biometric credential must be asymptotically separable.

□

We showed that *asymptotic separability* of the user and attacker distributions is a sufficient and necessary condition for enhancing the security of a wallet composed of a single binary credential with non zero probabilities $P_{\text{safe}}, P_{\text{leak}}$ or P_{loss} by integrating it with a biometric credential.

6.3 Intuitive Sufficient Condition

Definition 2. A biometric credential with user distribution fucntion $P_u(s)$ and attacker distribution function $P_a(s)$ is said to be *asymptotically dominated* if:

$$\forall c > 0 \; \exists T \text{ s.t.} : \forall x \geq T : p_a(x) \leq c\, p_u(x)$$

or

$$\forall c > 0 \; \exists T \text{ s.t.} : \forall x \leq T : p_u(x) \leq c\, p_a(x)$$

We will show that *asymptotically dominancy* of user and attacker distribution functions is a sufficient condition for enhancing a binary based wallet's security.

Proposition 4. *For any wallet w composed of one binary credential and success function $P^{binary}_{success}$, augmenting the wallet with an asymptotically dominated biometric credential improves its achievable security. That is,*

$$P^{integrated}_{success}(T) > P^{binary}_{success}.$$

Proof. A binary credential is characterized by fixed probability states $P^1_{\text{safe}}, P^1_{\text{loss}}, P^1_{\text{leak}}, P^1_{\text{theft}}$. Thus the success probability of a wallet composed solely of this binary credential is constant and given by:

$$P^{\text{binary}}_{\text{success}} = P^1_{\text{safe}}$$

A biometric credential is characterized by 4 probability states that depend on the operating threshold T: $P^2_{\text{safe}}(T), P^2_{\text{loss}}(T), P^2_{\text{leak}}(T), P^2_{\text{theft}}(T)$. We can describe each one of these probabilities as a function of False acceptance rate (FAR) and False rejection rate (FRR) as follows:

$$\begin{aligned}
P^2_{\text{safe}}(T) &= 1 - \text{FAR}(T) - \text{FRR}(T) + \text{FAR}(T)\text{FRR}(T), \\
P^2_{\text{loss}}(T) &= \text{FRR}(T)\big(1 - \text{FAR}(T)\big), \\
P^2_{\text{leak}}(T) &= \text{FAR}(T)\big(1 - \text{FRR}(T)\big), \\
P^2_{\text{theft}}(T) &= \text{FAR}(T)\text{FRR}(T).
\end{aligned}$$

The success probability of the integrated wallet comprised of 2 credentials is described by Equation 2 and Equation 3. Thus, we can express the success probability of the integrated wallet as a function of the operating threshold T as shown in Equation 4 and Equation 5. We previously observed that a sufficient condition for proving improvement of achievable security is described by Equation 6 and Equation 7. We need to show that there exists an operating threshold T such that either:

$$\frac{\text{FRR}(T)}{1 - \text{FAR}(T)} < \frac{P^1_{\text{leak}}}{P^1_{\text{safe}} + P^1_{\text{leak}}}$$

or

$$\frac{\text{FAR}(T)}{1 - \text{FRR}(T)} < \frac{P^1_{\text{loss}}}{P^1_{\text{safe}} + P^1_{\text{loss}}}$$

We will expand left hand side of Equation 6.

$$\frac{\mathrm{FRR}(T)}{1-\mathrm{FAR}(T)} = \frac{\int_{-\infty}^{T} P_u(s)ds}{\int_{-\infty}^{T} P_a(s)ds}$$

By Definition 2, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold T such that for every $s \leq T$, $P_u(s) \leq c \cdot P_a(s)$. Therefore:

$$\frac{\int_{-\infty}^{T} P_u(s)ds}{\int_{-\infty}^{T} P_a(s)ds} \leq \frac{\int_{-\infty}^{T} c \cdot P_a(s)ds}{\int_{-\infty}^{T} P_a(s)ds} = c$$

This is true $\forall c > 0$, thus we can choose c such that it satisfies Equation 6.
We will expand left hand side of Equation 7.

$$\frac{\mathrm{FAR}(T)}{1-\mathrm{FRR}(T)} = \frac{\int_{T}^{\infty} P_a(s)ds}{\int_{T}^{\infty} P_u(s)ds}$$

By Definition 2, since the biometric credential is asymptotically dominated, it follows that there exists an operating threshold T such that for every $s \geq T$, $P_a(s) \leq c \cdot P_u(s)$. Therefore:

$$\frac{\int_{T}^{\infty} P_a(s)ds}{\int_{T}^{\infty} P_u(s)ds} \leq \frac{\int_{T}^{\infty} c \cdot P_u(s)ds}{\int_{T}^{\infty} P_u(s)ds} = c, \forall c > 0$$

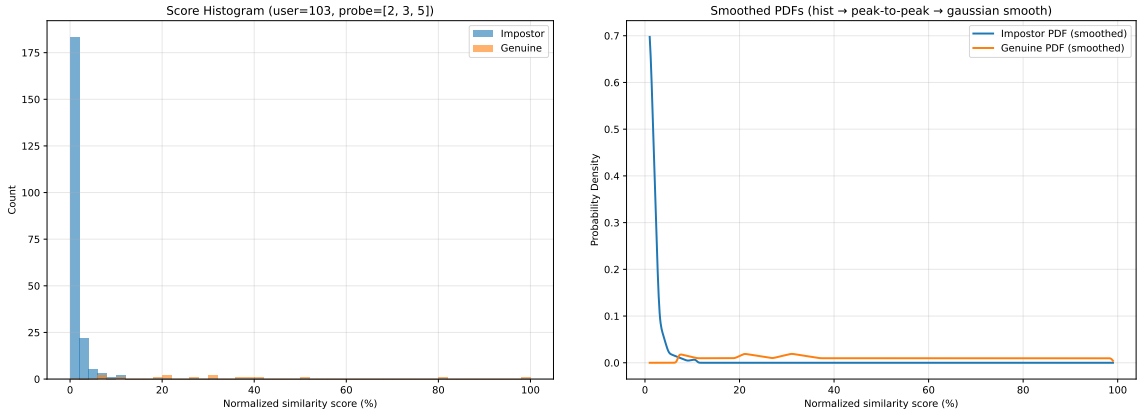
We showed that $\forall c > 0$ either $\frac{\mathrm{FAR}(T)}{1-\mathrm{FRR}(T)} < c$ or $\frac{\mathrm{FRR}(T)}{1-\mathrm{FAR}(T)} < c$. Therefore, integrating a binary based wallet with an asymptotically dominated biometric credential improves its security. □

This proof is not trivial, as the integration of another binary credential to a binary credential based wallet, does not always enhance its security. For example, if the wallet is comprised of 1 binary credential with high safe values and low leak values, adding a binary credential with low safe values and high leak values would decrease the overall security. Thus, Proposition 4 shows that by using an *asymptotically dominated* biometric credential, we can always improve the security of any binary credential based wallet. Moreover, *asymptotically dominancy* condition supports practicale scenarios, as we expect that for very low values of score s the attacker distribution dominates the user while vise versa for very high values of s. We will show measuremens supports this claim in Section 6.

7 Measuremens

We will now present measurements based on fingerprint taken from FVC2002 database [1] and FVC2004 database [2]. We used SourceAFIS an opensource algorithm created by Vazan [25] for scoring fingerprints out of their images based on minutiae triplets. We chose to use fingerprints, as it is a widely common and used example of a biometric credential in authentication mechanisms. For a test set, we took 3 templates out of FVC2002 database 1 of user 103 and compare them against the whole database of both FVC2002 and FVC2004. A geniune will be considered 3 templates of user 103 agains its other templates while imposter will be all the other combinations. We maintained same approach as Jain et al [10] and Cappelli et al [5] and first built histogram of the geniune and imposter combinations as shown in Figure 14a. Then, we turned the histogram into a smoothed probability distribution function as shown in Figure 14b. We found the False rejection rate and False acceptance rate and computed the Equal Error Rate (EER) as shown in Figure 15. The most important outcome from this approach was to achieve similar EER rate of around 1.8% as achieved from Maio et al. [13]

so we could objectively make a comparison. We computed the probability success function as given in Equation 1 of a wallet composed of a single continuous credential and showed that the optimal operating point is different from the EER point and achieves higher success probability as shown in Figure 16a. Finally, we computed the probability success function of a wallet composed of a single discrete credential with $P_{\text{safe}} = 0.9$ and $P_{\text{leak}} = 0.04$ $P_{\text{loss}} = 0.04$ and integrated it with the continuous credential.



(a) Histogram of the genuine and imposter combinations of the fingerprint scores

(b) Smoothed probability distribution function of the genuine and imposter combinations of the fingerprint scores

Figure 14: From histogram to probability distribution function

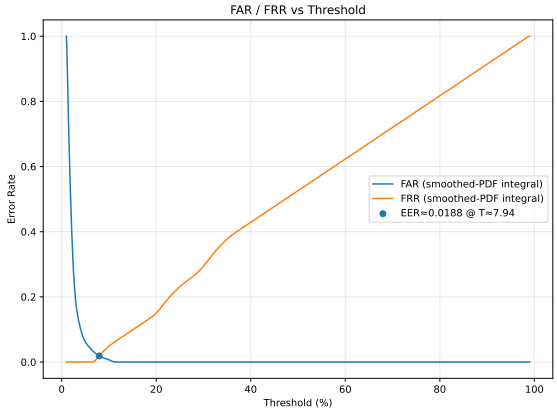
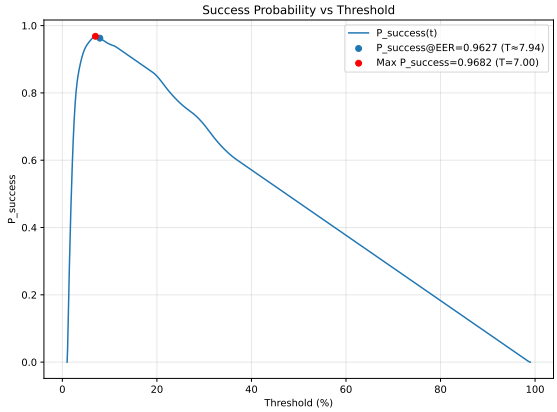
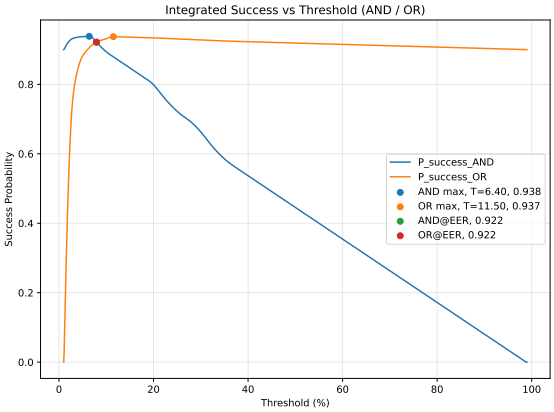


Figure 15: FAR vs. FRR curves for the fingerprint scores



(a) Success probability function of a wallet composed of a single continuous credential

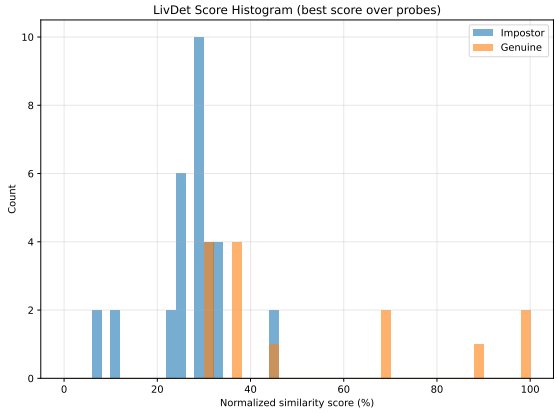


(b) Success probability function of a wallet composed of a single discrete credential and a single continuous credential

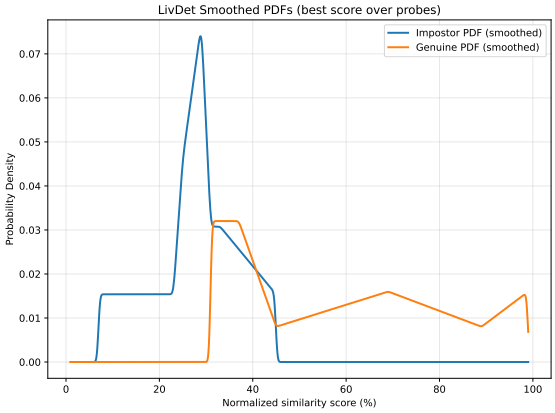
Figure 16: Continuous credential achieves higher success probability than the EER point in both continuous credential based wallet and integrated wallet

We showed that the optimal operating point of the integrated wallet is different from the EER point and achieves higher success probability by around 1.6% than the EER point, as the EER itself is only 1.8%. In addition, we can observe that the optimal point of the wallet achieves higher success probability than both the discrete only and continuous only wallets as shown in Figure 16b. Although there is an improvement in the success probability of the integrated wallet, we can observe the improvement is not as significant as the one observed in the theoretical analysis. This can be explained by the fact that the EER point of the continuous credential is already very low, thus there is not much room for improvement. In that scenario, we compared several templates of user 103 against templates of different users. However, our main approach in this paper was to increase the security level and success in scenarios where an imposter will try to spoof the user templates and authenticate into the user account. Hence, we expect the imposter to use spoofing tools, producing distribution function that much more similar to the genuine distribution function, which might lead to increased EER point.

To support this claim, we used the LivDet2015 database [8, 15, 18, 26], which contains a large-scale dataset of live and spoof fingerprint samples. We followed the same methodology as before and used the SourceAFIS algorithm [25]. As probe templates, we selected six Digital Persona scanner images of user 002_0. Each probe was matched against 14 additional images of the same user to generate genuine scores. For each target image, we define the final score as the maximum similarity across all probe templates. For example, if the matching scores between a target image and the six probes are [0.3, 0.5, 0.4, 0.8, 0.9, 0.7], then the resulting score is 0.9. To simulate a spoofing scenario, we used fake fingerprint images of the same user (002_0) produced using gelatine and matched them against the probe templates to obtain the imposter score distribution. The same maximum-score aggregation was applied in this case as well. We present the resulting histogram and the smoothed probability distribution functions in Figures 17a and 17b. We then computed the EER point and plotted the FAR-FRR curve, as shown in Figure 18. As expected, the EER increased significantly to 17.8%. Finally, we evaluated the success probability for both a single biometric credential, as shown in Figure 19a, and for a wallet composed of a single binary credential and a single biometric credential, as shown in Figure 19b. We can observe that the optimal operating point achieves a success probability approximately 14% higher than the EER point when integrating it with a binary credential with $P_{\text{safe}} = 0.8, P_{\text{loss}} = 0.04, P_{\text{leak}} = 0.15, P_{\text{theft}} = 0.01$ parameters, using the same algorithm.



(a) Histogram of the genuine and imposter combinations of the fingerprint scores



(b) Smoothed probability distribution function of the genuine and imposter combinations of the fingerprint scores

Figure 17: From histogram to probability distribution function

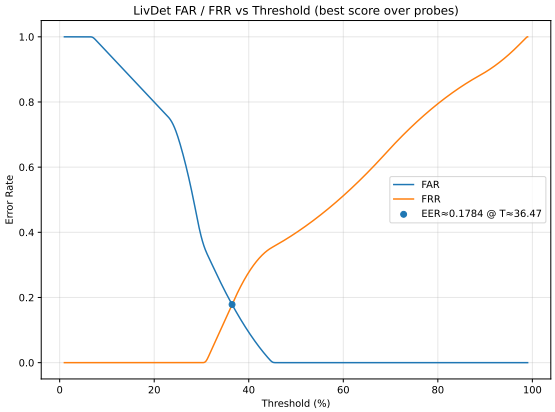
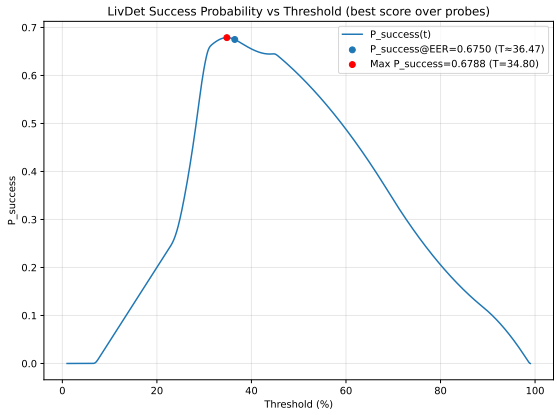
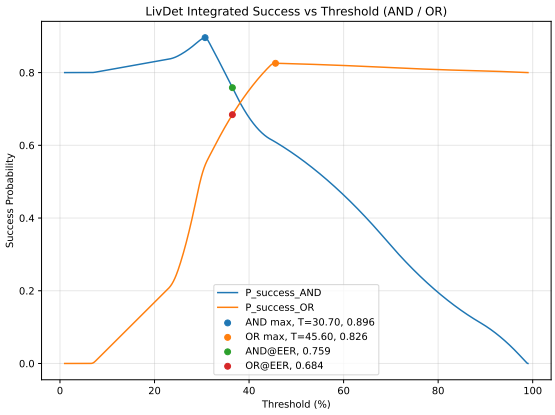


Figure 18: FAR vs. FRR curves for the fingerprint scores



(a) Success probability function of a wallet composed of a single biometric credential



(b) Success probability function of a wallet composed of a single biometric credential and a single binary credential with probabilities $P_{\text{safe}} = 0.8$, $P_{\text{loss}} = 0.04, P_{\text{leak}} = 0.15, P_{\text{theft}} = 0.01$.

Figure 19: Biometric credential achieves higher success probability than the EER point in both biometric credential based wallet and integrated wallet

8 Conclusion

We formalize the wallet authentication model consisting, for the first time, of both biometric alone and multi-credentials wallets. We analyze the security of each wallet type and present a new operating point that achieves higher success probability by modelling each credential with four probability states. We show every wallet comprised of binary credentials can be enhanced by integrating it with asymptotically separable biometric credential. Our analysis shows significant improvement of the success probability under spoofing attack scenarios, which became prevalant and easier on biometric systems in today's artificial intelligence era. Building new mechanisms, based on the suggested dynamic operating point done (§4) and stricting the threshold when adversaries are using spoofing tools to attack the biometric system is a promising direction for future work. Our findings have important and impactful implications for the design of secure authentication biometric systems. Harnessing our approach, will affect millions of users around the world and improve their privacy and security of their digital assets.

References

[1] Fingerprint verification competition 2002 (fvc2002). `http://bias.csr.unibo.it/fvc2002/`, 2002. Biometric Systems Laboratory, University of Bologna.

[2] Fingerprint verification competition 2004 (fvc2004). `http://bias.csr.unibo.it/fvc2004/`, 2004. Biometric Systems Laboratory, University of Bologna.

[3] Kumar Abhishek, Sahana Roshan, Prabhat Kumar, and Rajeev Ranjan. A comprehensive study on multifactor authentication schemes. In Natarajan Meghanathan, Dhinaharan Nagamalai, and Nabendu Chaki, editors, *Advances in Computing and Information Technology*, pages 561–568, Berlin, Heidelberg, 2013. Springer Berlin Heidelberg.

[4] Matt Bishop. *Introduction to Computer Security*. Addison-Wesley Professional, 2004.

[5] R. Cappelli, D. Maio, D. Maltoni, J.L. Wayman, and A.K. Jain. Performance evaluation of fingerprint verification systems. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 28(1):3–18, 2006.

[6] Ittay Eyal. On Cryptocurrency Wallet Design. In Vincent Gramoli, Hanna Halaburda, and Rafael Pass, editors, *3rd International Conference on Blockchain Economics, Security and Protocols (Tokenomics 2021)*, volume 97 of *Open Access Series in Informatics (OASIcs)*, pages 4:1–4:16, Dagstuhl, Germany, 2022. Schloss Dagstuhl – Leibniz-Zentrum für Informatik.

[7] Federal Trade Commission. Consumer sentinel network data book 2021. `https://www.ftc.gov/reports/consumer-sentinel-network-data-book-2021`, 2022.

[8] Luca Ghiani, David Yambay, Valerio Mura, Simona Tocco, Gian Luca Marcialis, Fabio Roli, and Stephanie Schuckcrs. Livdet 2013 fingerprint liveness detection competition 2013. In *2013 International Conference on Biometrics (ICB)*, pages 1–6, 2013.

[9] Lin Hong and Anil Jain. Integrating faces and fingerprints for personal identification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 20(12):1295–1307, 1998.

[10] A. Jain, Lin Hong, and R. Bolle. On-line fingerprint verification. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 19(4):302–314, 1997.

[11] A.K. Jain, A. Ross, and S. Prabhakar. An introduction to biometric recognition. *IEEE Transactions on Circuits and Systems for Video Technology*, 14(1):4–20, 2004.

[12] Ahmed Mahfouz, Rawad Abdulghafor, and Alaa A. K. Ismaeel. Passkeys in practice: An empirical evaluation of fido2/webauthn compliance and interoperability. In *2025 IEEE 24th International Conference on Trust, Security and Privacy in Computing and Communications (TrustCom)*, pages 2781–2788, 2025.

[13] D. Maio, D. Maltoni, R. Cappelli, J.L. Wayman, and A.K. Jain. Fvc2000: fingerprint verification competition. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 24(3):402–412, 2002.

[14] Deepak Maram, Mahimna Kelkar, and Ittay Eyal. Interactive multi-credential authentication. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, CCS '24, page 408–422, New York, NY, USA, 2024. Association for Computing Machinery.

[15] Gian Luca Marcialis, Aaron Lewicke, Bozhao Tan, Pietro Coli, Dominic Grimberg, Alberto Congiu, Alessandra Tidu, Fabio Roli, and Stephanie Schuckers. First international fingerprint liveness detection competition—livdet 2009. In Pasquale Foggia, Carlo Sansone, and Mario Vento, editors, *Image Analysis and Processing – ICIAP 2009*, pages 12–23, Berlin, Heidelberg, 2009. Springer Berlin Heidelberg.

[16] Robert Morris and Ken Thompson. Password security: a case history. *Commun. ACM*, 22(11):594–597, November 1979.

[17] Marwa Mouallem and Ittay Eyal. Asynchronous authentication. In *Proceedings of the 2024 on ACM SIGSAC Conference on Computer and Communications Security*, CCS '24, page 3257–3271, New York, NY, USA, 2024. Association for Computing Machinery.

[18] Valerio Mura, Luca Ghiani, Gian Luca Marcialis, Fabio Roli, David A. Yambay, and Stephanie A. Schuckers. Livdet 2015 fingerprint liveness detection competition 2015. In *2015 IEEE 7th International Conference on Biometrics Theory, Applications and Systems (BTAS)*, pages 1–6, 2015.

[19] S. Prabhakar, S. Pankanti, and A.K. Jain. Biometric recognition: security and privacy concerns. *IEEE Security & Privacy*, 1(2):33–42, 2003.

[20] Eva Reisinger, Lena Genthner, Jules Kerssemakers, Philip Kensche, Stefan Borufka, Alke Jugold, Andreas Kling, Manuel Prinz, Ingrid Scholz, Gideon Zipprich, Roland Eils, Christian Lawerenz, and Jürgen Eils. Otp: An automatized system for managing and processing ngs data. *Journal of Biotechnology*, 261:53–62, 2017. Bioinformatics Solutions for Big Data Analysis in Life Sciences presented by the German Network for Bioinformatics Infrastructure.

[21] S. Ribaric and I. Fratric. A biometric identification system based on eigenpalm and eigenfinger features. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 27(11):1698–1709, 2005.

[22] Arpita Sarkar and Binod K Singh. A review on performance, security and various biometric template protection schemes for biometric authentication systems. *Multimedia Tools and Applications*, 79(37):27721–27776, 2020.

[23] Fred B. Schneider. Authentication for people (draft textbook chapter). `https://www.cs.cornell.edu/fbs/publications/chptr.AuthPeople.pdf`, 2009. Retrieved October 2022.

[24] Shaun Cheon. What is biometric identity theft and what protections exist? `https://andopen.co.kr/what-is-biometric-identity-theft-and-what-protections-exist/`, 2025.

[25] Robert Vazan. SourceAFIS: A Fingerprint Recognition Algorithm. `https://sourceafis.machinezoo.com`, 2018.

[26] David Yambay, Luca Ghiani, Paolo Denti, Gian Luca Marcialis, Fabio Roli, and S Schuckers. Livdet 2011 — fingerprint liveness detection competition 2011. In *2012 5th IAPR International Conference on Biometrics (ICB)*, pages 208–215, 2012.